

(Tools (GUI Table, Normalization tool, AML).

Aim: To perform normalization upto BCNF Based on given dependencies.

Banking database:-

1. Identify Banking attributes:- Customer, Account, Branch, Banker info, loan, credit card.
2. Relational Schema: Banking (Customer, Account, Branch, Banker info, loan, credit card)
3. Functional dependencies (FD's between Attributes):

Customer ID $\rightarrow$ Name, Address, Ph-no

Account number $\rightarrow$ Account name, category.

Branch-ID $\rightarrow$ Branch Name, Location, ifsc-code.

Banker-ID $\rightarrow$ Banker name, Ph-no.

Customer-ID $\rightarrow$ Account-number

loan-ID $\rightarrow$ loan-Amount.

Customer-ID $\rightarrow$ loan-ID

Step 2:- Convert to 1NF:

- * No repeating groups or arrays
- * All attributes are atomic

The Schemas in 1NF.

Step 3:- Convert to 2NF

- * All primary keys are single-column keys, so no partial dependencies exist.
- * However, we ensure foreign key attributes are managed correctly.

output:- The schema is already in 2NF

Step 4:- Convert it to 3NF

Eliminate Transitive dependencies

* Customer-ID $\rightarrow$ Account-number $\rightarrow$ loan-ID
 $\rightarrow$ move loan-ID to a separate loan table.

* Customer-ID $\rightarrow$ Name, Address, Ph-NO
 $\rightarrow$ Already in separate users table.

* Account-number $\rightarrow$ Customer-ID $\rightarrow$ Branch-ID
 $\rightarrow$ No redundancy.

All transitive dependencies removed.

Step 5:- Convert to BCNF

check if every determinant is a candidate key:

* Customer-ID, Account-number, Branch-ID, loan-ID are all unique keys for their respective tables.

* foreign keys like Customer-ID... do not violate BCNF Rules.

All FD's comply with BCNF- no further decomposition needed.

Using Griffith Tool.

1. Input relational schema and functional dependencies.
2. Griffith tool generates a dependency graph
3. Analyze the graph to identify normalization issues.
4. Apply normalization to transform schema.
5. Verify the resulting schema meets BCNF criteria.

Griffiths 800 Step:-

1. Create a new project in Griffiths.
2. Define the relational schema and FD's
3. Run the "dependency Graph" tool
4. Analyze the graph for normalization issues
5. Apply Transformations using the 'Normalize' tool
6. verify BCNF compliance using "BCNF check" tool.

Normalization Schema:

Customer (Customer_ID, Name, Ph_no)

Account (Account-number, Account-name, Category)

Branch (Branch-DO, Branchname, Location, HSC-code)

Banker info (Banker_ID, Name, Ph_no)

loan (loan_ID, Customer-ID, Amount)

Credit_Card (Credit_Card-Number, Customer-ID, Limit)

VEL TECH	
EX No.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	5
VIVA VOCE (3)	4
RECORD (4)	14
TOTAL (15)	14
SIGN WITH DATE	15/10

Result:

Thus, the implementation of normalizing the database upto BCNF Based on given depending was successfully created.